ABINA K



Computer Science and Engineering Student
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LinkedIn | GitHub | LeetCode | HackerRank | CodeTantra SEA | CodeChef

EDUCATION

KPR Institute of Engineering and Technology, Coimbatore, Tamil Nadu — 2024 – 2028
B.E. in Computer Science and Engineering **CGPA: 8.4/10** — (up to 4th sem)

PKV Government Higher Secondary School, Mookkareddipatti, Dharmapuri — 2022 – 2024
HSC – Mathematics and Biology **88.67%**

PKV Government Higher Secondary School, Mookkareddipatti, Dharmapuri — 2021 – 2022
SSLC **89.2%**

PROJECTS

AQUA VELO — Water-Powered Car *STEM Project*

- Developed a model car that moves without fuel or electricity, using gravitational force and water pressure to convert the potential energy of stored water into kinetic and mechanical energy.
- Flowing water rotates a turbine that drives the vehicle forward, demonstrating basic hydrodynamics and energy conversion as a simple, eco-friendly concept for sustainable transportation.

INNOVSENSE — Alcohol Detection and Automatic Engine Cut-Off System

- Built a sensor-based system that detects alcohol in a driver's breath using an MQ-3 alcohol sensor; when the level exceeds a configured threshold, it automatically cuts off the engine and activates a buzzer to alert the driver.
- Combines embedded systems, sensor technology, a microcontroller platform and a MOSFET for automation and alerting, aiming to improve road safety and discourage drunk driving.

Smart Weight Monitoring in Public and Private Transport — AI/ML-Powered Overload Prevention System

- Monitors vehicle weight in real time using load cells and pressure sensors to identify potential overloading; AI/ML algorithms are intended to analyze sensor data, detect unsafe load conditions, and trigger alerts or automatically disable the engine when configured limits are exceeded.
- Integrates sensors, a microcontroller, Embedded C and Python, and IoT modules (Wi-Fi/GSM) to reduce accident risk and protect vehicle lifespan. Research paper on this project presented at CIT, Coimbatore.

TECHNICAL SKILLS

Programming Languages: C, Python, Java

Web Development: HTML, CSS

Platforms and Tools: GitHub, Google Colab, Kaggle, MATLAB, Tinkercad, HackerRank, CodeTantra SEA, LeetCode

CERTIFICATIONS

- **Python Libraries for Data Science** — Certification Course — Oct 21, 2025
- **Power BI for Beginners** — Certification Course — Jul 19, 2025
- **NPTEL** — Responsible and Safe AI Systems — Oct 2025
- **Software Engineer** — HackerRank — Dec 2025